IE802 Notes

Separation Problem



Figure 1. A squiggly "S" shape with "S not possible" below.

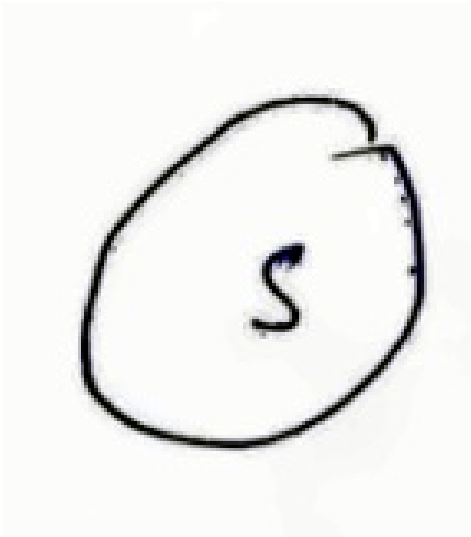


Figure 2. S and T separated by an arrow, labeled "Separate S from T by a hyperplane... on linear."

Separation Problem

relax into two parts



Figure 3. A wavy line graph with two labeled regions $S1$ (above) and $S2$ (below), with a red separating line through the middle.

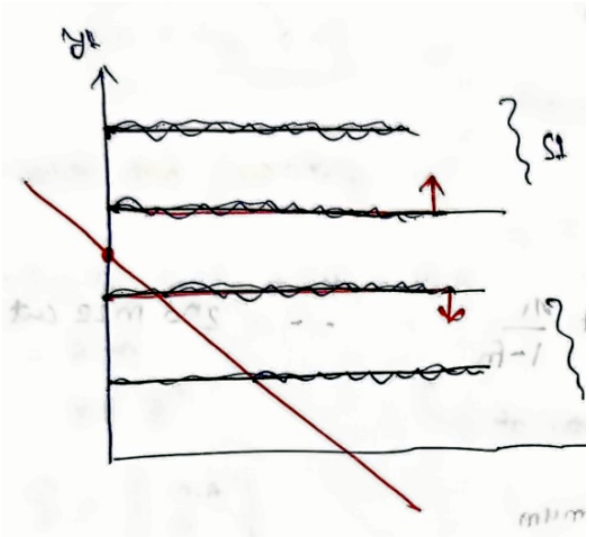


Figure 4. Two overlapping "S1" regions, each with a red separating line through them.

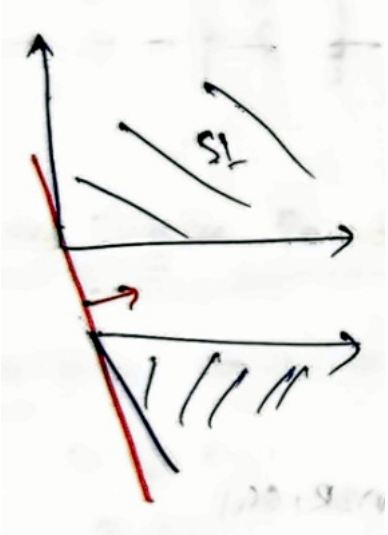


Figure 5. A scatter plot of points with a red separating line, labeled “minimize flat hat 2-norm” — “split cutg...” — “not necessary” — “no good” — “too big of separation” — “using convex hulls” — “shrink” — “and convex hull”.

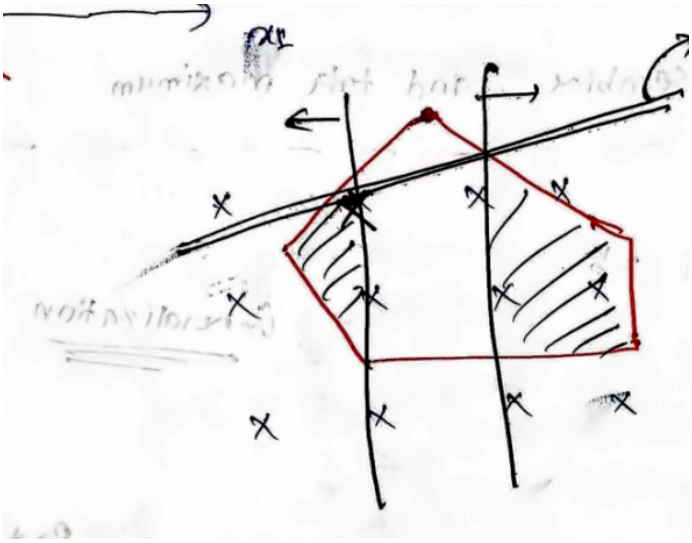


Figure 6. A polygonal region with a red separating line through it, labeled “split cutg...” — “this has been done before” — “(shrinkage needs more research)”.

Split inequalities:

If $\beta^* \leq \beta$

$\Rightarrow$ Then it is valid for S ($\forall x \in S$)



Figure 7. A set S split into two regions by a vertical line, labeled "S" on the left and "11" on the right.

$\hookrightarrow$ $\forall x \in B$: To check the inequality is valid for S, find $\beta^* = \max_{x \in S} \alpha x$ st. $x \in S$

Let $S = S_1 \cup S_2$

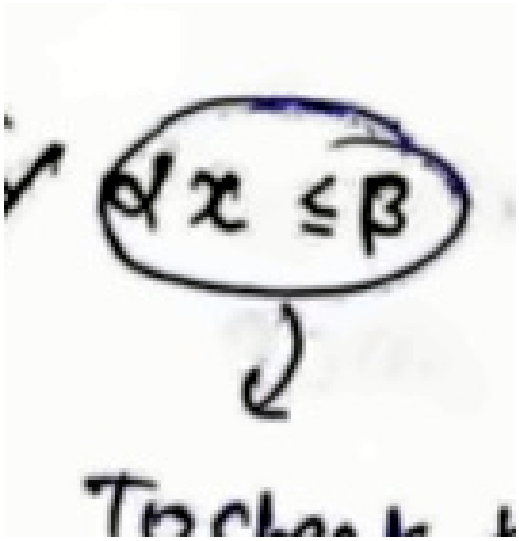


Figure 8. Two overlapping sets S_1 and S_2.

Suppose $\alpha x \leq \beta$ is valid for both S_1 and S_2.

Then $\alpha x \leq \beta$ is valid for S.

$\hookrightarrow$ how to find cutting planes?

$\hookrightarrow$ Split in different ways

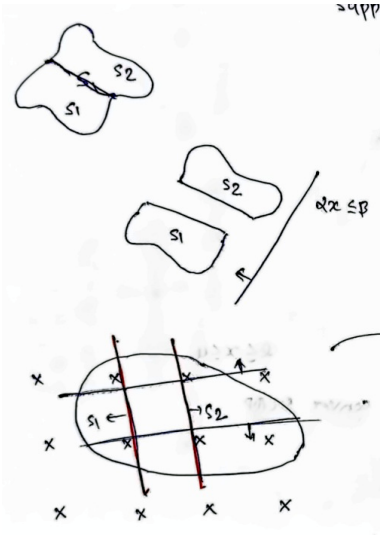


Figure 9. Two separate sets S_1 and S_2 each with a line $\alpha x \leq \beta$ through them.

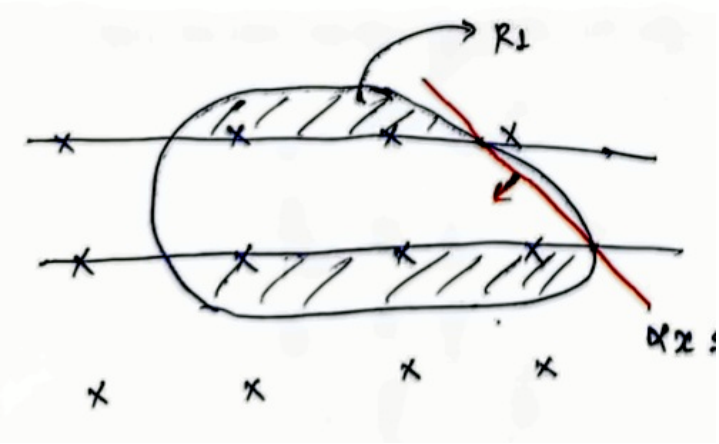


Figure 10. An oval region split by a red line into two halves, labeled R_1 on the left and a shaded region on the right.

↪ And inequality valid for both half

$$\begin{gathered}\min c^T x + d^T \theta \\ \text{st } g_i(x,\theta) \leq 0 \quad i = 1, 2 \dots m \\ x \in \mathbb{R}^n, \quad \theta \in \mathbb{Z}^p\end{gathered}$$

$$S = \{(x,\theta) : x \in \mathbb{R}^n, \theta \in \mathbb{Z}^p, g_i(x,\theta) \leq 0, i = 1, 2 \dots m\}$$

$$R = \{(x,y) : x \in \mathbb{R}^n, y \in \mathbb{R}^p, g_i(x,y) \leq 0, i = 1, 2 \dots m\}$$

Find (x, θ) st it is valid for S but not R

Step *Pick a disjunction (OR)*

- $y_1 \leq 0$ or $y_1 \geq 1$
- $y_5 \leq 4$ or $y_5 \geq 5$
- $y_5 + y_6 + y_3 \leq 1$ or $y_5 + y_6 + y_4 \geq 2$

$$S_1 = \{(x,y) : x \in \mathbb{R}^n, y \in \mathbb{Z}^p, g_i(x,y) \leq 0, i = 1, 2 \dots m, y_1 \leq 1\}$$
$$S_2 = \{\dots, y_1 \geq 2\}$$
$$S = S_1 \cup S_2$$

Step Relax $\rightarrow R_1$ and R_2

$$R_1 = \{(x,y) : x \in \mathbb{R}^n, y \in \mathbb{R}^p, \dots, y_1 \leq 1\}$$
$$R_2 = \{y \in \mathbb{R}^p, y_1 \geq 2\}$$

Convex Hull

Step (4) And $\alpha x + \gamma y \leq \beta$, valid for both R1 and R2

G use R1, R2 ... together to Red line. $\Rightarrow$ There are lots of lines like the Red line (infinite)

- Stuber & Mehrotra, 1999 Give the convex hull of $R_1 \cup R_2$ under the assumptions:

(i) y = binary
(ii) R_is are bounded
(iii) disjunctions used like $y_i = 0$ or $y_i = 1$

- Convex hull is given by $(x^1, y^1) \in R_1, (x^2, y^2) \in R_2$ $(x, y) = \lambda^1(x^1, y^1) + \lambda^2(x^2, y^2)$ with $\lambda^1 + \lambda^2 = 1, \lambda^1 \geq 0, \lambda^2 \geq 0$

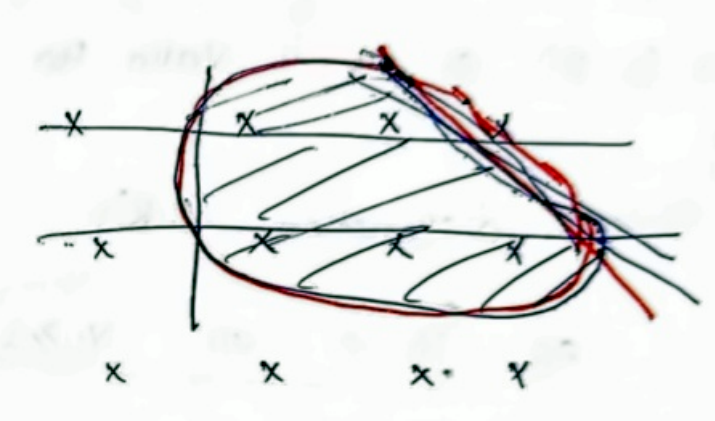


Figure 11. A red line (convex hull boundary) through a set of marked points on an oval curve, illustrating the convex hull concept.

$$M(x^1)$$

$$m = \begin{bmatrix} x_1^1 \\ x_1^2 \\ y_1^1 \\ y_1^2 \\ \lambda_1^1 \\ \lambda_1^2 \end{bmatrix}, \quad \text{assuming } y_1 = 0 \text{ or } y_1 = 1$$

Form new constraints in place of difficult set:

$$x = \bar{x}_1 + \bar{x}^2, \quad y = \bar{y}_1 + \bar{y}_2$$

difficult posn $\rightarrow$ base posn

$$\exists i(x^i, y^i) \leq 0 \rightarrow cx$$
$$\exists i(x^2, y^2) \leq 0 \rightarrow cx$$
$$x = \lambda^1 x^1 + \lambda^2 x^2 \rightarrow ncx$$
$$y = \lambda^1 y^1 + \lambda^2 y^2 \rightarrow ncx$$
$$\lambda^1 + \lambda^2 = 1 \rightarrow cx$$
$$x^1, x^2 \geq 0 \rightarrow cx$$
$$y_1^1 = 0, y_1^2 = 1 \rightarrow cx$$